Numerical Integration

Example Problem

Part 1. Integration from Datasets

Here, we DO NOT assume evenly distributed data

Estimate the velocity from the dataset of acceleration

```
clear all


x=[0 5 10 15 20 25 30 35 40 45 50 55 60];
y=[0 3 8 20 33 42 40 48 60 12 8 4 3];
N=length(x)
```

```
N =
    13
```

Matlab function for Discrete data: I_matlab = trapz(x,y);

```
formatSpec = 'I= %3.6f  with Error %2.3f%% \n';


% Matlab function
I_matlab = trapz(x,y);


I_true=I_matlab;
fprintf(formatSpec,I_matlab,0)
```

```
I= 1397.500000  with Error 0.000%
```

Rectangle Method:

```
% Rectangle Method
Ir=0;
for k=1:N-1
    Ir= Ir+y(k)*(x(k+1)-x(k));
end


error_Ir=abs((Ir-I_true)/I_true)*100;
fprintf(formatSpec,Ir,error_Ir)
```

```
I= 1390.000000  with Error 0.537%
```

Trapezoidal Method

```
It=0;
for k=1:N-1
```

```
    It=It+ 0.5* (y(k)+y(k+1))*(x(k+1)-x(k));
end


error_It=abs((It-I_true)/I_true)*100;
fprintf(formatSpec,It,error_It)
```

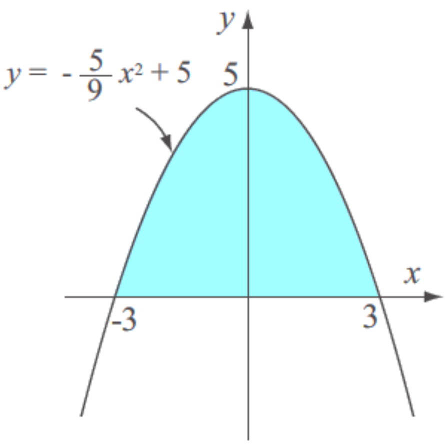
```
I= 1397.500000  with Error 0.000%
```

Part 2. Integration from a Function

Here, we assume evenly distributed data

Example Problem

The area of the shaded region shown in the figure can be calculated by:



$$A = \int_{-3}^{3} \left(-\frac{5}{9}x^2 + 5\right) \, dx = 20$$

```
%% Example 1
clear, clc
format long
a=-3; b=3;
N=8;
h=(b-a)/N;
x=a:h:b;


fun=@(x) -(5/9)*x.^2+5;
I_true=20;
```

Example 2: Area of Unit Semi-Circle

```
a=-1; b=1;
N=8;
h=(b-a)/N;
fun=@(x) (1-x.^2).^0.5;
I_true=pi/2;
```

Evaluate and compare the numerical integral using the several methods: when N=4, N=8

```
formatSpec = 'I= %3.6f  with Error %2.3f%% \n';


% Matlab function
I_matlab = quad(fun,a,b);
error_matlab=abs((I_matlab-I_true)/I_true)*100;
fprintf(formatSpec,I_matlab,error_matlab)
```

```
I= 20.000000  with Error 0.000%
```

```
% Rectangle Method
Ir = IntegrateRectangle(fun,a,b,N);
error_Ir=abs((Ir-I_true)/I_true)*100;
fprintf(formatSpec,Ir,error_Ir)
```

```
I= 19.687500  with Error 1.563%
```

```
% Trapezoidal Method
It=trapezoidal(fun,a,b,N);
error_It=abs((It-I_true)/I_true)*100;
fprintf(formatSpec,It,error_It)
```

```
I= 19.687500  with Error 1.563%
```

```
% Simpson 1/3 Method
Is=Simpson(fun,a,b,N);
error_Is=abs((Is-I_true)/I_true)*100;
fprintf(formatSpec,Is,error_Is)
```

```
I= 20.000000  with Error 0.000%
```

```
% Adaptive Simpson Method
Ia = AdaptSimpson(fun,a,b);
error_Ia=abs((Ia-I_true)/I_true)*100;
fprintf(formatSpec,Ia,error_Ia)
```

```
I= 20.000000  with Error 0.000%
```

Lecture Objective

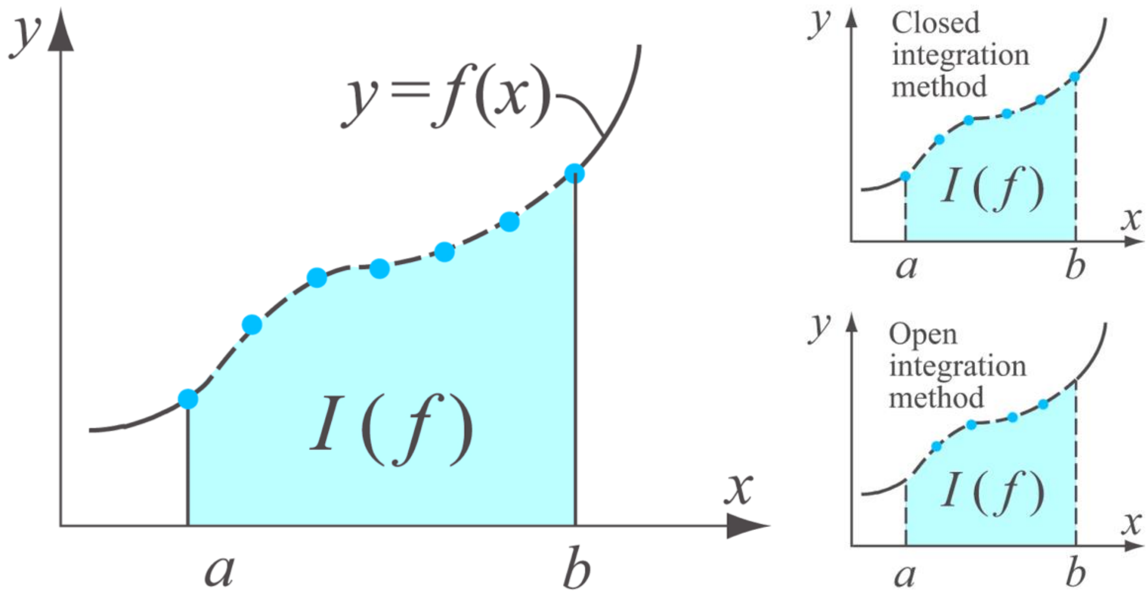
This lecture will cover how to calculate a numerical approximation of the integration from (1) discrete datasets (2) a function.

$$I(f) = \int_a^b f(x)dx$$

The numerical integration methods can be divided by

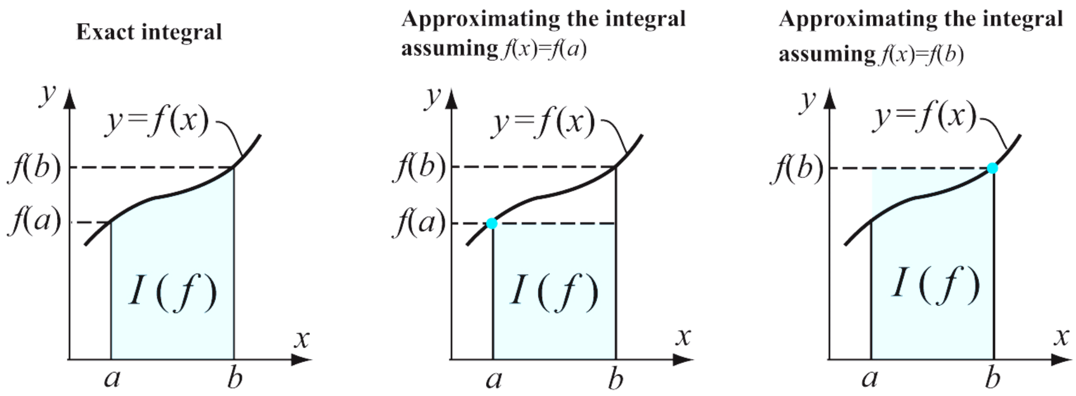
- **Closed methods**
  - End- points of each intervals are used
  - **Rectangular, Trapezoidal, Simpson Methods**
- Open Methods
  - Intervals of integration extend beyond the range of the end-points
  - **Midpoint**, Gauss Quadrature methods

There are other options such as apply interpolation or curve-fitting on the discrete dataset and integrate analytically.



Rectangular, Midpoint, Trapezoidal Methods

The simplest approximation of the integration is to take f(x) over the interval as a simple rectangle



For $I(f)$ of an rectangle area: *We will use $I(f)$ *as* the <u>numerical</u> Integration

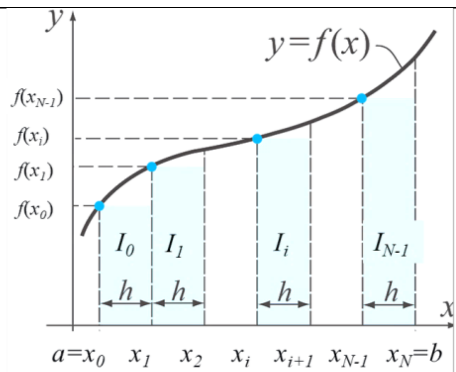
$$I(f) = \int_a^b f(a)dx = f(a)(b-a) \quad \text{or} \quad I(f) = \int_a^b f(b)dx = f(b)(b-a)$$

Composite Rectangular Method O(h)

The domain [a, b] are divided by N subintervals (i.e., N rectangles) the whole integration is adding the area of each rectangle.

#Intervals: N #points= N+1 = m Intervals in the range of (x[0] to x[N])

$$I = f(x_0)(x_1-x_0) + f(x_1)(x_2-x_1) + \cdots + f(x_{N-1})(x_N-x_{N-1})$$



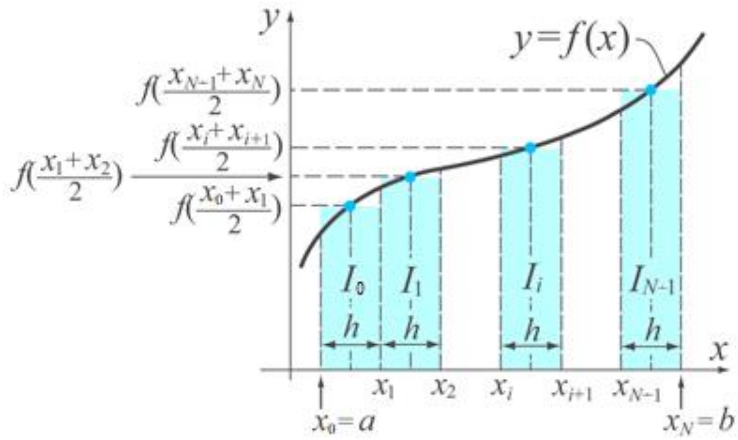
$$\therefore I = \sum_{k=0}^{N-1} f(x_k)\underbrace{(x_{k+1}-x_k)}$$

↳ can be substituted as h if it is a constant.

Composite Mid-Point Method $O(h^2)$

The height of each sub-intervals is obtained from the mid point.

#Intervals: N   #points= N+1   Intervals in the range of (x[0] to x[N])



$$I = f\left(\frac{x_1+x_0}{2}\right)(x_1-x_0)$$

$$+ f\left(\frac{x_2+x_1}{2}\right)(x_2-x_1) + f\left(\frac{x_3-x_2}{2}\right)(x_3-x_2)$$

$$+ \cdots + f\left(\frac{x_N+x_{N-1}}{2}\right)(x_N-x_{N-1})$$

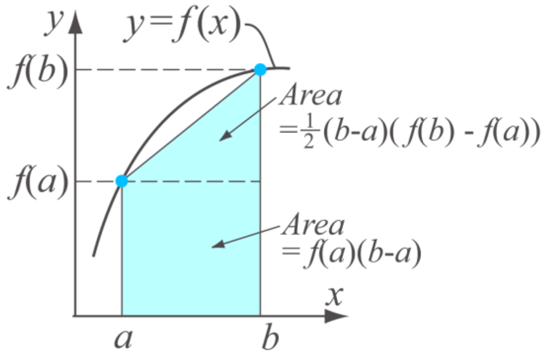
$$\therefore I = \sum_{k=0}^{N-1} f\left(\frac{x_{k+1}+x_k}{2}\right)(x_{k+1}-x_k)$$

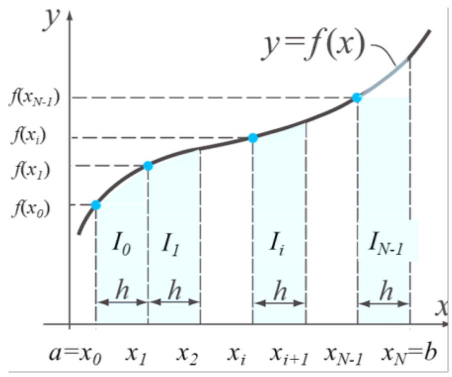
$(x_{k+1}-x_k)$ can be substituted as h if it is a constant.

Composite Trapezoidal Method O(h^2)

The area of each sub-intervals is obtained from the trapezoid

#Intervals: N   #points= N+1   Intervals in the range of (x[0] to x[N])





$$I = \frac{1}{2}\Big(f(x_1)+f(x_0)\Big)(x_1-x_0) + \frac{1}{2}\Big(f(x_2)+f(x_1)\Big)(x_2-x_1)$$

$$+ \cdots + \frac{1}{2}\Big(f(x_N)+f(x_{N-1})\Big)(x_N-x_{N-1})$$

$$\therefore I = \frac{1}{2}\sum_{k=0}^{N-1}\Big(f(x_{k+1})+f(x_k)\Big)(x_{k+1}-x_k)$$

If $(x_{k+1}-x_k)$ is a constant h,

$$I = \frac{h}{2}\sum_{k=0}^{N-1}\Big(f(x_{k+1})-f(x_k)\Big)$$

Simpson's Methods

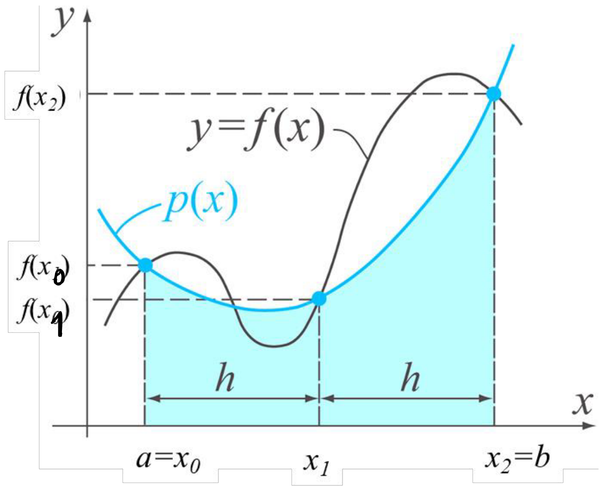
The trapezoidal for each subinterval is *interpolation* with a line (1st order polynomial). For a better results, we can apply the interpolation with quadratic(2nd order) or cubic(3rd order) to approximate the integrals.

- Interpolate with 1st polynomial: Trapezoidal Method $O(h^2)$
- Interpolate with 2nd polynomial: Simpson 1/3 Method $O(h^4)$
- Interpolate with 3rd polynomial: Simpson 3/8 Method $O(h^4)$

Simpson 1/3 Method $O(h^4)$

We need 3 points for unique quadratic polynomial.

Create the 3rd point as the mid-point between [a,b]



Write a quadratic equation in the form of Newton's polynomial: 3 unknowns using x[0], x[1], x[2]. Let h=(x1-x0) and 2*h=(x2-x0)

$$p(x) = a_0 + a_1(x - x[0]) + a_2(x - x[0])(x - x[1])$$

$$P(x_0) = a_0 = f(x_0)$$

$$P(x[0]) = a_0 = f(x_0)$$

$$a_1 = \frac{f(x[1]) - f(x[0])}{x[1] - x[0]} = \frac{f(x[1]) - f(x[0])}{h} = \frac{f(x_1) - f(x_0)}{h}$$

What are the solutions for the 3 unknowns (a0, a1, a2) ? Write the full formula of the quadratic polynomial
Then Integrate it from x[0] to x[2].

$$a_2 = \frac{\dfrac{f(x[2]) - f(x[1])}{x[2]-x[1]} - \dfrac{f(x[1]) - f(x[0])}{x[1]-x[0]}}{x[2]-x[0]} = \frac{\dfrac{f(x_2)-f(x_1)}{h} - \dfrac{f(x_1)-f(x_0)}{h}}{2h}$$

$$= \frac{\dfrac{f(x_2)-f(x_1)-f(x_1)+f(x_0)}{h}}{2h}$$

$$= \frac{f(x_2)-2f(x_1)+f(x_0)}{2h^2}$$

$$\therefore P(x) = f(x_0) + \overbrace{\frac{f(x_1)-f(x_0)}{h}}^{a_1 \, \shortparallel}(x - x[0]) + \overbrace{\frac{f(x_2)-2f(x_1)+f(x_0)}{2h^2}}^{a_2 \, \shortparallel}(x-x[0])(x-x[1])$$

Let $x[0] = x_0$. $x[1] = x_1$, $x[2] = x_2$

$x - x_0 = u$ (substitution)

$dx = du$

$$x - x_1 = x - (x_0 + h)$$
$$= u - h$$

$$\int_0^{2h} P(u+x_0) = \int_0^{2h} f(x_0) + \frac{f(x_1)-f(x_0)}{h} u + \frac{f(x_2)-2f(x_1)+f(x_0)}{2h^2} \overbrace{u(u-h)}^{u^2 - hu} du$$

$$= 2hf(x_0) + \frac{f(x_1)-f(x_0)}{h} \cdot \frac{1}{2}(2h)^2 + \frac{f(x_2)-2f(x_1)+f(x_0)}{2h^2}\left[\frac{1}{3}u^3 - \frac{h}{2}u^2\right]_0^{2h}$$

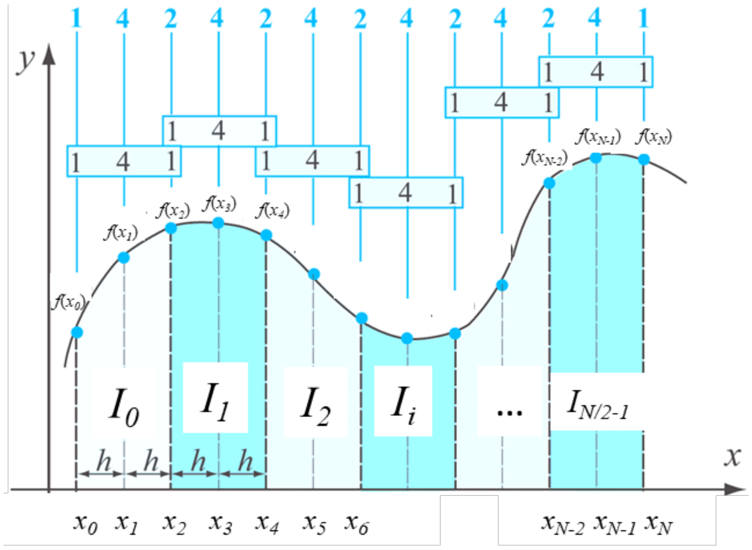
$$= 2hf(x_0) + \frac{f(x_1)-f(x_0)}{\cancel{h}} \cdot 2h^{\cancel{2}} + \frac{f(x_2)-2f(x_1)+f(x_0)}{2h^2} \cdot \left(\frac{1}{3}\cdot 8h^3 - \frac{h}{\cancel{2}} \overset{2}{\cancel{4}} h^2\right)$$

$$= 2hf(x_1) + \frac{f(x_2)-2f(x_1)+f(x_0)}{\cancel{2h^2}} \cdot \frac{\cancel{2}}{3} h^{\cancel{3}}$$

$$= \left(2 - \frac{2}{3}\right) f(x_1) + \frac{f(x_2)+f(x_0)}{3} \cdot h = \boxed{\frac{h}{3}\left(f(x_2) + 4f(x_1) + f(x_0)\right)}$$

Composite Simpson 1/3 formula

Divide into ***N*** *equally spaced* subinvervals. Here, **N** must be even numbers (we have used two subintervals for each sub-integration)



$$I_0 = \frac{h}{3}\left(f(x_0) + 4f(x_1) + f(x_2)\right)$$

$$I_1 = \frac{h}{3}\left(f(x_2) + 4f(x_3) + f(x_4)\right)$$

$$\vdots$$

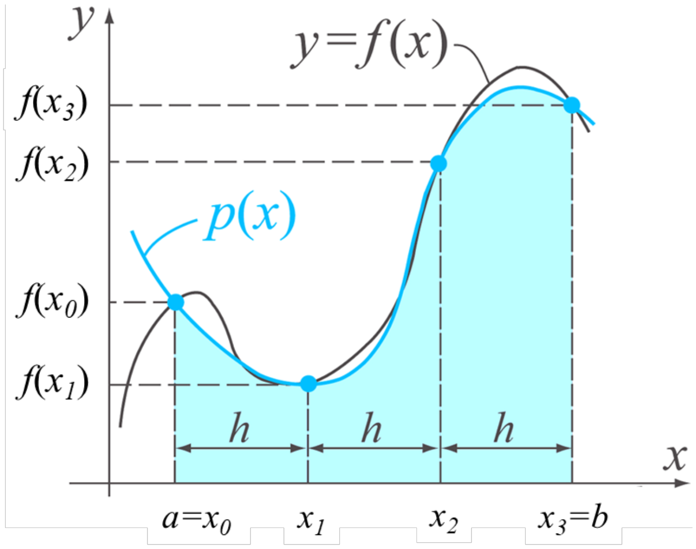
$$I_{\frac{N}{2}-1} = \frac{h}{3}\left(f(x_{N-2}) + 4f(x_{N-1}) + f(x_N)\right)$$

$$\therefore I = \frac{h}{3}\left(f(x_0) + 4f(x_{N-1}) + f(x_N)\right) + \frac{h}{3}\sum_{k=1}^{\frac{N}{2}-1}\left(4f(x_{2k-1}) + 2f(x_{2k})\right)$$

Simpson 3/8 Method O(h^4)

We need 4 points for unique cubic polynomial.

Create the 2 more points between [a,b] as



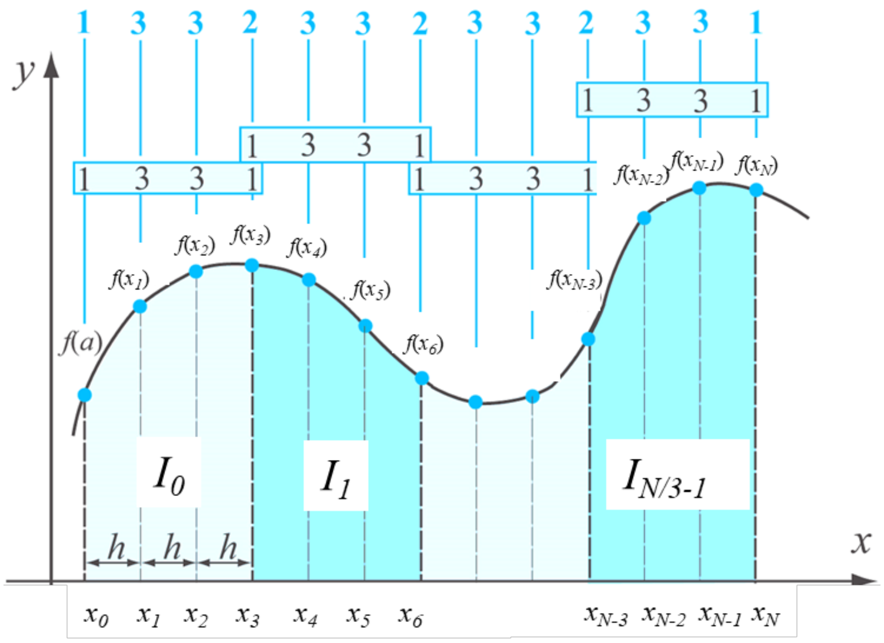
Similar Process as Simpson 3/8, it gives the integration as

$$I_0 = \int_{a=x_0}^{b=x_3} p(x)dx = \frac{3}{8}h[f(x_0)+3f(x_1)+3f(x_2)+f(x_3)]$$

Composite Simpson 3/8 formula

Divide into ***N*** *equally spaced. But N should be divisible by 3*

We can combine 1/3 and 3/8 formula for odd number of N.



$$I_0 = \frac{3}{8}h\left[f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)\right]$$

$$I_1 = \frac{3}{8}h\left[f(x_3) + 3f(x_4) + 3f(x_5) + f(x_6)\right]$$

$$I_2 = \frac{3}{8}h\left[f(x_6) + 3f(x_7) + 3f(x_8) + f(x_9)\right]$$

$$\vdots$$

$$I_{\frac{N}{3}-2} = \frac{3}{8}h\left[f(x_{N-6}) + 3f(x_{N-5}) + 3f(x_{N-4}) + f(x_{N-3})\right]$$

$$I_{\frac{N}{3}-1} = \frac{3}{8}h\left[f(x_{N-3}) + 3f(x_{N-2}) + 3f(x_{N-1}) + f(x_N)\right]$$

$$I = \frac{3}{8}h\left[f(x_0) + f(x_N)\right] + \frac{3}{8}h \times 3 \sum_{k=1}^{\frac{N}{3}} \left(f(x_{3k-2}) + f(x_{3k-1})\right)$$

$$+ \frac{3}{8}h \times 2 \sum_{k=1}^{\frac{N}{3}-1} f(x_{3k})$$

$$= \frac{3}{8}h\left[f(x_0) + 3f(x_{N-2}) + 3f(x_{N-1}) + f(x_N)\right] + \frac{3}{8}h \sum_{k=1}^{\frac{N}{3}-1} \left(3f(x_{3k-2}) + 3f(x_{3k-1}) + 2f(x_{3k})\right)$$

Summary

For N intervals, x[0] to x[N]

Composite Rectangular Method O(h)

Different intervals	Same interval *h*
$I(f)=\sum_{i=0}^{N-1} f(x_i)(x_{i+1}-x_i)$	$I(f)=h\sum_{i=0}^{N-1} f(x_i)$

Composite Mid-Point Method O(h^2)

Different intervals	Same interval h
$I(f)=\sum_{i=0}^{N-1} f\left(\frac{x_i+x_{i+1}}{2}\right)(x_{i+1}-x_i)$	$I(f)=h\sum_{i=0}^{N-1} f\left(\frac{x_i+x_{i+1}}{2}\right)$

Composite Trapezoidal Method O(h^2)

Different intervals	Same interval *h*
$I(f)=\frac{1}{2}\sum_{i=0}^{N-1}\left[f\left(x_i\right)+f\left(x_{i+1}\right)\right](x_{i+1}-x_i)$	$I(f)=\frac{h}{2}\sum_{i=0}^{N-1}\left(f\left(x_i\right)+f\left(x_{i+1}\right)\right)$ $=\frac{1}{2}\left(f\left(x_0\right)+f\left(x_N\right)\right)+h\sum_{i=0}^{N-1}\left(f\left(x_i\right)+f\left(x_{i+1}\right)\right)$

Composite Simpson 1/3 Method O(h^4)

For N even numbers, same intervals

$$I=\frac{h}{3}[f(x_0)+4\sum_{i=1,3,5}^{N-1} f\left(x_i\right)+2\sum_{k=2,4,6}^{N-2} f\left(x_k\right)+f(x_N)]$$

Composite Simpson 3/8 formula

For N multiple of 3 and, same intervals

$$I=\frac{3h}{8}\left[f(x_0)+3\sum_{i=1,4,7}^{N-2}\left[f\left(x_i\right)+f\left(x_{i+1}\right)\right]+2\sum_{k=3,6,9}^{N-3} f\left(x_k\right)+f(x_N)\right]$$